# Grandmothers' care and mothers' time: coresident grandmothers in India

**Abstract**

Do coresident grandmothers relieve mothers of childcare, and what do they give up? India's Time Use Survey 2024 records same-day diaries for every household member aged six and above. I compare 49,992 married mothers of children under six, 36% of whom live with a grandmother, and study 26,062 women and 19,382 men aged 45 to 80 in the generation above young married women. The roster omits each child's mother, so I assign children and grandparents by generation. Older women living with a child under six report 79.5 minutes more childcare a day and less domestic work, free time and sleep. Mothers' childcare and paid-work minutes differ little with and without a grandmother and change with the specification; their employment rate is 1.4 percentage points lower. Grandmother care is associated with a large time cost for older women and, in the pooled sample, with no more paid work by mothers.

**Keywords:** grandparents · childcare · three-generation households · patrilocality · time use · India

**JEL Classification:** D13 · J13 · J22

## 1 Introduction

Grandparents are a major source of care for young children in many societies, and caring for grandchildren is a large part of the daily life of many older women. In India the family is the main childcare institution. Public provision for children under six runs mainly through the *anganwadi* centers of the Integrated Child Development Services, which focus on supplementary nutrition, health and pre-school education rather than full-day care, and women's labor force participation has long been low by international standards and slow to rise with education and income (Klasen and Pieters 2015; Afridi et al. 2018; Chatterjee et al. 2018; Fletcher et al. 2019). A common presumption, in policy discussion and in the family itself, is that coresident grandparents, above all grandmothers, absorb much of the care of young children and so free mothers to work, rest or invest in other children. The presumption has empirical support elsewhere: in the United States, Europe, China and Japan the availability of grandparents is associated with, and in several quasi-experimental designs causes, higher maternal employment (Ogawa and Ermisch 1996; Sasaki 2002; Maurer-Fazio et al. 2011; Posadas and Vidal-Fernández 2013; Arpino et al. 2014; Compton and Pollak 2014; Bratti et al. 2018).

The same arrangement has a second side that matters for the economics of aging and of the family. The time that grandmothers give to grandchildren is a transfer from a generation whose own work, rest and health also matter, in a country whose older population is growing (International Institute for Population Sciences et al. 2020; United Nations Population Fund and International Institute for Population Sciences 2023). There are also reasons to doubt that the evidence from other settings carries over. Residence after marriage is predominantly patrilocal, so the coresident grandmother is typically the mother-in-law, and the literature describes the young daughter-in-law as the junior member of a hierarchy in which her mobility and work outside the home are shaped by her husband's parents (Dyson and Moore 1983; Kandiyoti 1988; Jayachandran 2015; Anukriti et al. 2020). Women in joint households in rural India are less likely to be employed than women in nuclear households (Dhanaraj and Mahambare 2019). A three-generation household is also a larger household, with more meals to cook. Coresidence can therefore add to a mother's work as well as bring a potential substitute for her childcare. I call these the *substitution* and *constraint* channels.

Which dominates is hard to learn from existing data. Surveys of labor supply record whether a grandparent lives nearby and whether the mother works, but not what either does with the day. Surveys of grandparents record whether they look after grandchildren, but not the mother's time on the same day. The literature has therefore inferred substitution from labor-market outcomes, and the cost of care to grandmothers has rarely been measured outside high-income settings.

This paper uses India's Time Use Survey 2024 (National Statistics Office 2025), which records a 24-hour diary, coded to the International Classification of Activities for Time-Use Statistics 2016 (ICATUS) (United Nations Statistics Division 2016), for every household member aged six and above. Diaries of mothers and of the grandparents who live with them are kept for the same day, and the household roster's relationship codes identify, for each married woman, the generation above her. The identification rules matter. For a woman who is a married child of the head, or the spouse of one, the generation above is the head and the head's spouse, which is the typical patrilocal arrangement. And in a household where a son and his wife live with his parents and a child under six is present, the head's wife is usually that child's grandmother, not its mother: I exclude such women from the mother sample (2,189 of 52,181 married women aged 18 to 49 in households with a child under six) and study them as grandmothers.

The analysis has four aims. The first is to compare the time use of mothers of children under six with and without a coresident grandmother, across the whole day. The second is to ask whether three-generation households provide the same childcare more cheaply or more care in total. The third is to estimate how a young grandchild is associated with the time use of grandmothers and grandfathers, which shows the cost of care to those who give it. The fourth is to ask which grandmothers matter (by age and employment), how the association differs across household arrangements, and whether it vanishes where it should, among mothers whose youngest child is of school age.

Each estimate below is reported with its 95% confidence interval (CI). The pooled association between a coresident grandmother and the mother's primary childcare is small: -2.1 minutes (CI

-4.6 to 0.4). The association with the mother's paid work, including travel, is -0.4 minutes (CI -3.8 to 3.0), and her probability of being employed by usual status is 1.4 percentage points lower (CI -2.3 to -0.4). The minute estimates are fragile. The childcare estimate ranges from -13.2 to 8.3 minutes across the specifications in the robustness table, and subgroup estimates differ in size and, for paid work, in sign. The grandmother's side is much clearer: in households with a child under six, the grandmother or grandmothers together report 85.9 minutes (SE 1.1) of childcare a day, 37% of the combined childcare of mother and grandmother, and where a child under six is present they report 79.5 minutes more (CI 77.4 to 81.6) primary childcare than other older women. They also report less domestic work, free time and sleep. Summed over all household members aged 15 and above, primary childcare is 88.6 minutes higher (CI 84.0 to 93.2) in households with a grandmother.

The paper contributes to three literatures. For the economics of grandparental childcare, it replaces inference from employment with same-day time measures for both generations, and shows that the association with mothers' time appears to depend on the grandmother's age and employment and on the household arrangement. For research on population aging and older women, it measures directly the time that older women give to grandchildren in a patrilocal, middle-income setting and the other parts of their day from which it comes. For the study of family structure and women's work in South Asia, it documents relationship-based rules for assigning young children and grandparents in rosters that do not record parent identifiers. The estimates are conditional associations, not causal effects: living arrangements are chosen jointly with fertility, work and care, and the survey is a single cross-section.

## 2 Background and conceptual framework

### 2.1 Grandparental care and maternal employment

A large literature finds that the availability of grandparents raises mothers' labor supply. In the United States, Compton and Pollak (2014) show that married women who live close to their mothers or mothers-in-law have higher labor force participation, and Posadas and Vidal-Fernández (2013) find that grandparent-provided childcare raises mothers' participation. In Europe, Dimova and Wolff (2011) show that downward time transfers, mainly grandchild care, raise maternal labor supply, Arpino et al. (2014) find a positive effect of grandparental support in Italy, and Bratti et al. (2018) exploit Italian pension reforms that raised retirement ages, and so reduced grandmothers' availability, and find lower participation among their daughters. On the grandmothers' side, Frimmel et al. (2022) show with Austrian administrative data that a first grandchild raises the probability that a grandmother leaves the labor market, and Igel and Szydlik (2011) show that welfare-state arrangements shape the form of grandparental care across Europe. The evidence from Asia is closer to the Indian setting because coresidence, not proximity, is the usual arrangement. Ogawa and Ermisch (1996) and Sasaki (2002) find that coresidence with parents or parents-in-law raised Japanese married women's employment, and Maurer-Fazio et al. (2011) find a similar

association in urban China. For the United States, Bansak et al. (2026) find that living in an intergenerational household is positively related to employment among Asian women and negatively among white women.

Two features of this literature motivate the present paper. Almost all of it measures the consequence of grandparental care (maternal employment) rather than the care itself, and it comes mainly from settings in which coresidence or proximity is not tied to a strong patrilocal hierarchy. Where the grandmother is the mother-in-law and the head of the household, her presence may bring authority over the daughter-in-law's time as well as help with it.

### 2.2 Older women's time, aging and intergenerational transfers

Grandparental childcare is a transfer of time between generations. Economic models distinguish altruistic from exchange motives for transfers (Cox 1987; Altonji et al. 1997), and Cardia and Ng (2003) show that grandparents' time is an important input into childcare. Albertini et al. (2007) document that European regimes differ in the frequency and intensity of time and money transfers across generations. Time transfers have an opportunity cost for the giver: Rupert and Zanella (2018) find that the arrival of grandchildren lowers grandmothers' labor supply in the United States, and Lumsdaine and Vermeer (2015) that a new grandchild raises the probability that women retire. The evolutionary literature on grandmothering makes the same point from another angle, treating grandmothers' investment in grandchildren as a central and costly part of human life histories (Hawkes et al. 1998; Sear and Mace 2008). Time diaries for the older generation allow this cost to be measured directly, alongside the benefit to the mother, and allow it to be related to the grandmother's age and employment, which proxy her capacity to give care.

### 2.3 Patrilocality and women's work in India

India differs from the settings above in ways that could reverse the sign of the association between coresidence and mothers' work. Dyson and Moore (1983) argued that the kinship system of northern India, with village exogamy and patrilocal residence, limits women's autonomy, and Kandiyoti (1988) describes a patriarchal bargain in which young brides accept subordination in their husband's household in exchange for later authority as mothers-in-law. Anukriti et al. (2020) show that coresident mothers-in-law restrict daughters-in-law's mobility and social networks in rural Uttar Pradesh; status concerns can lead households to withdraw women from work outside the home (Eswaran et al. 2013), and social norms are a barrier to women's employment more broadly (Field et al. 2021; Jayachandran 2021). Dhanaraj and Mahambare (2019) find that women in nuclear households in rural India are more likely to work than women in joint households, and Afridi et al. (2018) decompose the fall in rural married women's participation over two decades. These studies observe employment, autonomy or decision-making, not time.

### 2.4 Measuring childcare in time diaries

Time-diary research shows that childcare is the most gendered of household tasks and that its measurement is sensitive to definitions (Sayer et al. 2004; Bianchi et al. 2006; Hook 2006; Craig and Mullan 2011). Primary childcare understates the time parents are responsible for children, because supervision is often recorded as a secondary activity or not at all (Folbre et al. 2005; Folbre 2006; Craig 2007; Kalenkoski and Foster 2008). I therefore report active and passive primary care separately, and childcare recorded as a secondary activity, because a grandmother may take over supervision while the mother keeps the care that requires her, or the reverse.

### 2.5 Conceptual framework

A mother allocates her day among childcare, domestic work, paid work and leisure. A coresident grandmother changes three things. She can supply childcare, which lowers the care the mother must provide (the substitution channel); this effect should rise with the grandmother's capacity, which declines with age and ill health and falls when she works herself. She raises household size and so domestic workload, and may need care herself (a workload channel). And in a patrilocal household she may raise the cost to the household of the daughter-in-law working outside the home (a norms channel); I group the last two as the constraint channel. Four predictions follow. If substitution is present, mothers' childcare is lower with a grandmother, more so for passive minding than for active care. Whether released time reaches paid work depends on the constraint channel, and if it is strong, released time shows up in domestic work or leisure. The grandmother's own time should show the other side, with more childcare and less of something else when a young grandchild is present. And substitution should be larger with grandmothers who have more capacity (younger, not employed) and concentrated where the youngest child is under six. The framework does not model the choice of living arrangement, the main threat to interpretation discussed below.

## 3 Data and methods

### 3.1 The Time Use Survey 2024

The Time Use Survey (TUS) 2024 was conducted by the National Statistics Office, Ministry of Statistics and Programme Implementation, between January and December 2024 (National Statistics Office 2025). It covers a nationally representative sample of households in rural and urban India, drawn in a stratified multi-stage design with First Stage Units (FSU), that is, villages and urban blocks, as primary sampling units. Every member aged six and above reports a 24-hour diary for the day before the interview, from 4 a.m. to 4 a.m., in time slots that can contain several activities. Each activity is coded to ICATUS 2016 and flagged as major or minor (secondary). I use every activity record: 10,211,478 records for 450,457 diarists in 139,487 households and 9,969 FSUs. Minutes of major activities are summed to the person-day, and all but five of the 450,457 diaries sum to exactly 1,440 minutes. I exclude 49 persons recorded as third gender from the anal-

yses because the sample is too small for separate estimates; they remain in household composition. Weights are the survey multipliers, and standard errors are clustered on the FSU (Solon et al. 2015; Abadie et al. 2023). The survey construction checks and the sample construction are in Online Resource 1.

### 3.2 Identifying mothers, children and grandparents

Children under six do not keep diaries but are listed on the household roster, so the number of under-six members is a direct count of roster members aged under six (the file contains members of all ages). As a check, the roster count of household members equals the household size recorded in the household file for 100.0% of households. The roster does not record the identifier of each child's mother. The candidate mothers are married women aged 18 to 49 whose relationship to the head is head or spouse of head (head level), or married child or spouse of a married child (junior level). Women with other relationships, such as a brother's wife, are excluded because the roster does not identify the generation above them. In a household in which a married child of the head lives with a spouse, a child under six is far more likely to be the junior couple's child, that is, the head's grandchild, than a late-born child of the head. I therefore exclude from the mother sample head-level women who live with a junior-level married woman aged 15 to 49 (2,189 women), and study them as grandmothers. The analysis sample consists of the 49,992 remaining women in households with at least one child under six. Of these, 33% (weighted) are at junior level, and among junior-level mothers 95% are daughters-in-law of the head. The husband is linked to 83% of mothers through the roster; I keep mothers whose husbands live elsewhere in the sample, with an explicit indicator. A mother's own children aged six to 17, a control below, are also assigned by roster position: for a head-level mother they are the head's children of that age, and for a junior-level mother the head's grandchildren of that age, but only where the household contains a single child-in-law of the head. Otherwise the count is set to missing and replaced by zero with an indicator for an unknown count; the youngest own child used for the placebo sample follows the same rule.

For a head-level mother, the generation above her consists of members coded as father, mother, father-in-law or mother-in-law of the head. For a junior-level mother it consists of the head and the head's spouse aged 40 or above; the lower limit admits young grandmothers while excluding heads' spouses who are likely to be the mother's sisters-in-law rather than her parents-in-law. A coresident grandmother is at least one woman in this generation, and a coresident grandfather at least one man. Because each mother is matched to the specific members of the generation above her, a grandmother is never the mother herself (zero self-links). By this definition 36% (weighted) of mothers of children under six live with a grandmother and 28% with a grandfather; 4.5% live with a grandfather but no grandmother. Among mothers with a coresident grandmother, 83% are at junior level, so most three-generation households are of the patrilocal type in which the young family lives in the older generation's household. The median age of the youngest coresident grandmother is 57 years, 60% of grandmothers are under 60, and 30% report employment as their

usual principal activity status (counts and weighted shares in Table S2 of Online Resource 1).

### 3.3 Time outcomes

Childcare is ICATUS 411 to 419 plus accompanying own children (442). Active care (411 to 415) covers physical care, medical care, teaching, talking and reading, and playing; passive care (416) is minding children. I also measure childcare recorded as a secondary activity, unpaid domestic services (division three) and within it food and meals (311 to 319), care of dependent adults (421 to 429 and 443), employment including related travel (division one), free time (divisions seven and eight) and sleep (911 to 919). Employment status is the usual principal activity status.

Table 1 compares the raw time use of mothers with and without a coresident grandmother. Mothers without a grandmother report 146.1 minutes (SE 1.0) of primary childcare a day and mothers with one 144.8 minutes (SE 1.1). The raw comparison mixes the grandmother's presence with everything that differs between joint and nuclear households, which motivates the regression approach.

Table 1: Mothers of children under six: minutes per day by grandmother coresidence

| Outcome | No grandmother | Grandmother present |
|---|---|---|
| Childcare (primary) | 146.1 (1.0) | 144.8 (1.1) |
| Active childcare | 141.2 (0.9) | 139.8 (1.1) |
| Passive childcare (minding) | 3.6 (0.2) | 3.5 (0.2) |
| Childcare as secondary activity | 45.6 (1.0) | 33.6 (0.8) |
| Unpaid domestic work | 358.7 (1.3) | 340.9 (1.6) |
| of which food and meals | 253.8 (1.1) | 242.0 (1.3) |
| Care of dependent adults | 0.1 (0.0) | 0.4 (0.1) |
| Employment (incl. travel) | 48.1 (1.2) | 49.7 (1.4) |
| Free time | 182.6 (1.1) | 193.5 (1.4) |
| Sleep | 527.9 (1.0) | 530.6 (1.0) |
| Employed (usual status, share) | 0.176 (0.004) | 0.165 (0.004) |

*Notes:* Married women aged 18–49 at head or junior level (head or spouse of head; married child of head or that child's spouse) in households with a child under six, excluding head-level women who live with a junior married woman of childbearing age. Survey-weighted means; FSU-clustered standard errors in parentheses. Employed is a share (three decimals).

### 3.4 Empirical strategy

For mother $i$ in household $h$, state $s$ and diary day $d$, I estimate

$$y_{ihsd} = \beta\, GM_h + X_{ih}'\gamma + \mu_s + \tau_d + \varepsilon_{ihsd}, \tag{1}$$

where $y_{ihsd}$ is a time outcome in minutes per day (or employment status) and $GM_h$ indicates a coresident grandmother. $X_{ih}$ includes the mother's age band and education, the husband's education and usual employment (with a separate category for mothers without a linked husband), the household's per-capita consumption quintile, urban residence, the number of children under six

and aged six to 14, the number of the mother's own children aged six to 17, caste group, religion, whether any member needs special care, and whether the diary was reported by a proxy. State fixed effects absorb regional kinship systems and labor markets, and day-of-week fixed effects absorb the weekly cycle. I do not control for household size in the main specification, because a grandmother raises it mechanically and the workload that comes with a larger household is part of the constraint channel; a robustness check adds it. I report 95% confidence intervals (CI) based on the normal approximation, not significance tests. In the text, an interval attached to a difference refers to the signed coefficient, with households without a grandmother as the reference.

The main threat to interpretation is selection into coresidence. Three-generation households differ from nuclear ones in wealth, caste, region and norms, and coresidence can respond to care needs: a grandmother may move in because a grandchild is born, or because she needs care herself. I read $\beta$ as a conditional association and assess it in four ways. First, I compare arrangements selected for different reasons. Junior-level mothers live in the older generation's household, the default patrilocal arrangement; head-level mothers whose parents or in-laws live with them have taken the older generation in, which may more often reflect the grandparents' need for support. Most junior-level mothers live with a grandmother (2,054 do not, against 15,196 who do), so the grandmother indicator is closely tied to the household arrangement, and where either cell has fewer than 30 mothers I do not report the estimate. Second, I examine gradients that the substitution channel predicts but selection on fixed household traits does not obviously predict, by the grandmother's age, employment and sex. Third, I estimate the specification for mothers whose youngest child is aged six to 14 and who have no child under six in the household, among whom substitution of childcare should be weaker. Fourth, I report the bias-adjusted coefficient of Oster (2019; see also Altonji et al. 2005).

The older generation's time is estimated on a separate sample of 26,062 women and 19,382 men aged 45 to 80 in the generation above a married woman of childbearing age. I regress the same outcomes on an indicator for a child under six in the household, with controls for age band, education, marital status, urban residence, consumption quintile, children aged six to 14, caste, religion, special care needs and proxy reporting, and state and day fixed effects. The comparison is between older-generation members whose grandchildren are young and those whose grandchildren are older or not yet born. Because every member of this sample already lives with a married woman of childbearing age, the comparison turns mainly on the timing of the junior couple's fertility. Fertility timing and living arrangements can still be chosen together, for example when a couple stays in the parental household because a child is born, so these estimates are also conditional associations. The older-generation sample starts at 45, while the grandmother link for junior-level mothers starts at 40. Online Resource 1 (Table S3) re-estimates the older women's equations from age 40, and the robustness table defines the grandmother as a woman aged 45 or above.

### 3.5 Ethics, data availability and the use of artificial intelligence tools

The analysis uses anonymized unit-level data released by the National Statistics Office and involved no contact with respondents; no ethical review was required. The microdata are available from the Ministry's microdata portal subject to its terms of use; analysis code will be made available on acceptance. Artificial intelligence (AI) tools (Claude (Anthropic) and other AI assistants) were used to help write and check analysis code, to check references and to draft text. The author reviewed and verified all code, results, references and text and takes full responsibility for the content.

## 4 Results

### 4.1 Mothers' time

Table 2 reports the main estimates. A coresident grandmother is associated with 2.1 minutes less (CI -4.6 to 0.4) primary childcare by the mother a day. Relative to the mean of mothers without a grandmother, 146.1 minutes, the point estimate is 1% of the baseline. The interval includes zero: the data are compatible with a reduction of up to 3% of the baseline and with no change. The split by type of care shows -2.2 minutes (CI -4.6 to 0.3) for active care and -0.1 minutes (CI -0.6 to 0.5) for passive minding. The intervals for both components include zero, and passive minding, the component the first prediction expects to fall most, is almost unchanged, so that prediction is not borne out in primary care. Mothers with a coresident grandmother do record less childcare as a secondary activity (-9.6 minutes (CI -11.8 to -7.5)), so supervision is lower only when it is recorded as a secondary activity, not as primary minding.

Table 2: Coresident grandmother and mothers' time use

| Outcome | Mean | Coefficient | 95 percent CI | N |
|---|---|---|---|---|
| Childcare (primary) | 146.1 | -2.09 | [-4.58, 0.40] | 49,992 |
| Active childcare | 141.2 | -2.17 | [-4.60, 0.25] | 49,992 |
| Passive childcare (minding) | 3.6 | -0.05 | [-0.56, 0.47] | 49,992 |
| Childcare as secondary activity | 45.6 | -9.60 | [-11.75, -7.46] | 49,992 |
| Unpaid domestic work | 358.7 | 1.14 | [-2.19, 4.47] | 49,992 |
| of which food and meals | 253.8 | 5.01 | [2.34, 7.67] | 49,992 |
| Care of dependent adults | 0.1 | -0.08 | [-0.23, 0.06] | 49,992 |
| Employment (incl. travel) | 48.1 | -0.41 | [-3.79, 2.97] | 49,992 |
| Free time | 182.6 | 4.34 | [1.39, 7.30] | 49,992 |
| Sleep | 527.9 | -3.10 | [-5.30, -0.91] | 49,992 |
| Employed (usual status, share) | 0.176 | -0.014 | [-0.023, -0.004] | 49,992 |

*Notes:* Each row is a separate regression of the outcome (minutes per day, or a 0/1 indicator for usual employment) on an indicator for a coresident grandmother and the full control set, with state and day-of-week fixed effects. Mean: outcome mean for mothers without a coresident grandmother. Survey-weighted; FSU-clustered standard errors; 95 percent confidence intervals use the normal approximation.

The remaining rows show where the mother's day goes. Unpaid domestic work differs by 1.1

minutes (CI -2.2 to 4.5), but time on food and meals, the largest domestic task and the one most directly raised by household size, is higher by 5.0 minutes (CI 2.3 to 7.7), an interval that excludes zero. Care of dependent adults differs by -0.1 minutes (CI -0.2 to 0.1). Employment including travel differs by -0.4 minutes (CI -3.8 to 3.0), and the probability that the mother's usual status is employed is 1.4 percentage points lower (CI -2.3 to -0.4). Neither measure shows more paid work with a grandmother present, and the employment-rate estimate points to a small reduction, a pattern that is consistent with the constraint channel in patrilocal households although these data do not isolate it. Free time is 4.3 minutes higher (CI 1.4 to 7.3) and sleep 3.1 minutes lower (CI -5.3 to -0.9). Both intervals exclude zero, but the differences amount to a few minutes in a day of 1,440.

Taken together, the pooled estimates show small differences in mothers' time, no sign that a grandmother's presence raises mothers' paid work, and an association with childcare whose interval includes zero. They do not show that grandmothers relieve mothers of care, and, as the robustness checks and subgroup results below show, they hide differences by grandmother and by household arrangement.

### 4.2 Who provides childcare in three-generation households

Table 3 (and Fig. S1 in Online Resource 1) decompose the childcare provided by household members. In households with a grandmother, the grandmother or grandmothers in the household together provide 85.9 minutes (SE 1.1) of childcare a day. They also report 148.7 minutes (SE 1.6) of domestic work and 98.9 minutes (SE 2.1) of employment (lower panel of Table 3), which shows what else their day contains. Summed over all members aged 15 and above, raw childcare is 324.4 minutes with a grandmother and 201.4 without, a gap of 123.0 minutes. Part of this gap reflects the larger number of children in three-generation households: conditional only on the numbers of children under six and aged six to 14 and on state and day fixed effects, the gap is 115.4 minutes (CI 110.5 to 120.3), and with the full control set it is 88.6 minutes (CI 84.0 to 93.2). Recorded adult childcare time is thus higher in three-generation households. Part of the difference may reflect two adults recording care of the same child at the same time, and part the care of several grandchildren at once; the diary does not identify the recipient. Among mothers with a linked husband, husbands' childcare averages 36.8 minutes with a grandmother and 48.3 without, and the adjusted difference is -14.1 minutes (CI -15.6 to -12.6). Grandmothers' care is therefore associated with less care by fathers as well as with more care in the household as a whole.

### 4.3 The older generation's time

Table 4 (and Fig. S2 in Online Resource 1) turns to the providers of care. Among older-generation women, a child under six in the household is associated with 79.5 minutes more (CI 77.4 to 81.6) primary childcare a day, against a mean of 4.0 minutes for women without a young grandchild. On the care side the third prediction of the framework holds: grandmothers report more childcare when

Table 3: Childcare minutes per day by provider, households with a child under six

| Provider | No grandmother | Grandmother present | Adjusted difference [95% CI] |
|---|---|---|---|
| Mother | 146.1 (1.0) | 144.8 (1.1) | -2.09 [-4.58, 0.40] |
| Husband (zero if not linked) | 44.3 (0.6) | 24.4 (0.5) | -12.48 [-13.78, -11.17] |
| Linked husbands only | 48.3 (0.6) | 36.8 (0.6) | -14.10 [-15.62, -12.59] |
| Grandmother(s), together | 0.0 | 85.9 (1.1) | 85.50 [83.39, 87.62] |
| Grandfather(s), together | 3.2 (0.2) | 18.0 (0.5) | 11.30 [10.27, 12.33] |
| Other members aged 15+ | 7.8 (0.3) | 51.3 (1.4) | 6.39 [4.87, 7.90] |
| All members aged 15+ | 201.4 (1.4) | 324.4 (2.7) | 88.62 [84.04, 93.20] |
| conditional on numbers of children only | | | 115.37 [110.45, 120.29] |
| *Grandmother(s), other activities* | | | |
| Unpaid domestic work | – | 148.7 (1.6) | |
| Employment (incl. travel) | – | 98.9 (2.1) | |

*Notes:* Unit: mothers in the analysis sample (49,992). Columns 1–2: minutes of primary childcare of each household member, survey-weighted means with FSU-clustered standard errors in parentheses. Column 3: coefficient on the grandmother indicator [95 percent confidence interval] from the main specification, with that row as the outcome; the linked-husband row is estimated on mothers with a linked husband (41,447). Husbands who are not linked through the roster count as zero in the husband row and their care falls in Other members. The conditional row controls only for the numbers of children under six and aged 6–14 and state and day fixed effects. All members is the household total and includes care given to other children in the household; its raw difference is 123.0 minutes. Lower panel: minutes of the grandmother(s) in households with a grandmother.

a young grandchild is present. The associations with grandmothers' domestic work (-16.6 minutes (CI -20.5 to -12.7)), paid work (4.7 minutes (CI -0.1 to 9.6)), employment status (1.6 percentage points higher (CI 0.4 to 2.8)), free time (-53.3 minutes (CI -58.0 to -48.5)) and sleep (-9.2 minutes (CI -12.0 to -6.4)) indicate where that time comes from. Grandmothers with a young grandchild report less domestic work, free time and sleep, with confidence intervals that exclude zero. Lower domestic work need not be a cost to them: mothers in three-generation households spend more time on food and meals (Table 2), which is consistent with domestic tasks moving to the younger woman, although the two estimates come from separate samples. The cost of the care they provide falls on their free time and sleep. The fall in free time is the largest, at 14% of the mean for women without a young grandchild. Their paid work does not fall, so in this sample the cost falls on free time and sleep rather than on earnings.

For older-generation men, the corresponding childcare association is 29.8 minutes (CI 28.3 to 31.2), and the women-minus-men difference is 49.7 minutes (CI 47.2 to 52.3). The care response of the older generation falls overwhelmingly on grandmothers, mirroring the gender division of care in the middle generation. Paid work of older men differs by 11.8 minutes (CI 3.7 to 19.9). It is higher for older men and not clearly for older women, but the women-minus-men difference, -7.1 minutes (CI -16.5 to 2.4), is imprecise and includes zero, so the data do not establish a gender difference in the paid-work response.

### 4.4 Which grandmothers matter

Table 5 examines the fourth prediction. Grandmother age proxies for health and stamina, which the survey does not measure directly. Relative to mothers without a grandmother, the coefficient

Table 4: The older generation: time use and a child under six in the household

| Outcome | Mean, women | Older women | Older men | Difference |
|---|---|---|---|---|
| Childcare (primary) | 4.0 | 79.48 [77.37, 81.60] | 29.76 [28.30, 31.22] | 49.72 [47.15, 52.29] |
| Active childcare | 3.4 | 70.43 [68.40, 72.45] | 26.86 [25.47, 28.26] | 43.57 [41.11, 46.03] |
| Passive childcare (minding) | 0.5 | 7.65 [6.82, 8.47] | 2.42 [2.00, 2.84] | 5.23 [4.30, 6.16] |
| Unpaid domestic work | 149.7 | -16.58 [-20.45, -12.70] | -1.22 [-3.05, 0.61] | -15.35 [-19.64, -11.07] |
| Employment (incl. travel) | 74.5 | 4.73 [-0.11, 9.57] | 11.80 [3.71, 19.88] | -7.07 [-16.49, 2.36] |
| Employed (usual status, share) | 0.219 | 0.016 [0.004, 0.028] | 0.026 [0.012, 0.040] | -0.010 [-0.029, 0.008] |
| Free time | 392.5 | -53.25 [-58.04, -48.46] | -33.54 [-39.69, -27.40] | -19.71 [-27.50, -11.92] |
| Sleep | 595.8 | -9.19 [-11.99, -6.39] | -6.16 [-9.48, -2.85] | -3.03 [-7.37, 1.31] |

*Notes:* Members aged 45–80 in the generation above a married woman of childbearing age (26,062 women, 19,382 men). Estimate [95 percent confidence interval] for an indicator for a child under six in the household, separately for women and men; Difference: women minus men, with a standard error that ignores the covariance of the two estimates (conservative, since grandmothers and grandfathers often share a household). Mean: women without a child under six. Controls: age band, education, marital status, urban, consumption quintile, children 6–14, caste, religion, special care need, proxy report; state and day fixed effects. FSU-clustered standard errors; 95 percent confidence intervals use the normal approximation.

on mothers' childcare is 3.1 minutes (CI 0.1 to 6.1) for a grandmother under 60 and -8.8 minutes (CI -11.8 to -5.8) for a grandmother aged 60 or older; the difference is 11.9 minutes (CI 8.5 to 15.2). The reduction in mothers' childcare is concentrated where the grandmother is 60 or older, and the interval for the difference excludes zero. This runs against a simple capacity story in which younger grandmothers are fitter, but it is consistent with older grandmothers being less likely to be in paid work and so more available at home.

Table 5: Mothers' childcare and employment minutes by type of coresident grandparent

| Comparison | Childcare | Employment |
|---|---|---|
| Grandmother under 60 | 3.06 [0.06, 6.06] | -3.52 [-7.50, 0.47] |
| Grandmother 60 or older | -8.80 [-11.77, -5.82] | 3.64 [-0.72, 7.99] |
| Difference (under 60 minus 60+) | 11.86 [8.49, 15.22] | -7.16 [-11.99, -2.32] |
| Grandmother not employed | -5.24 [-7.93, -2.56] | -2.75 [-6.48, 0.99] |
| Grandmother employed | 5.82 [2.04, 9.60] | 5.46 [-0.01, 10.93] |
| Difference (not employed minus employed) | -11.06 [-14.93, -7.20] | -8.21 [-14.07, -2.34] |
| Grandmother only | -8.89 [-12.08, -5.69] | 9.77 [4.58, 14.95] |
| Grandfather only | -5.76 [-10.25, -1.27] | -6.73 [-12.78, -0.67] |
| Both grandparents | 1.12 [-1.89, 4.13] | -8.13 [-11.91, -4.35] |
| Difference (grandmother only minus grandfather only) | -3.13 [-8.07, 1.82] | 16.50 [8.99, 24.01] |

*Notes:* Each block is one regression per outcome with mutually exclusive indicators; the omitted group is mothers with no coresident grandmother (first two blocks) or no coresident grandparent (third block). Age is that of the youngest coresident grandmother; employment is usual principal activity status. Controls as in the main specification. Estimate [95 percent confidence interval]; FSU-clustered.

An employed grandmother has less time to give. The coefficient is -5.2 minutes (CI -7.9 to -2.6) for a grandmother who is not employed and 5.8 minutes (CI 2.0 to 9.6) for one who is, a difference of -11.1 minutes (CI -14.9 to -7.2). The lower childcare is concentrated among mothers whose grandmother is not in paid work; with an employed grandmother mothers' childcare is higher. Grandmothers' age, employment and household position are related, and employment is itself a choice made jointly with the care they provide, so these comparisons are descriptive.

The last block separates grandmothers from grandfathers. The coefficient on mothers' childcare is -8.9 minutes (CI -12.1 to -5.7) for a grandmother without a grandfather, -5.8 minutes (CI -10.3 to -1.3) for a grandfather without a grandmother, and 1.1 minutes (CI -1.9 to 4.1) for both together. With both grandparents present mothers' childcare is not lower, so the pooled reduction comes from households with a single coresident grandparent. The interval for the difference between grandmothers and grandfathers includes zero; households with only a grandfather are few, so this comparison is imprecise. Mothers' paid work is 9.8 minutes higher (CI 4.6 to 15.0) with a grandmother only and 8.1 minutes lower (CI -11.9 to -4.4) with both grandparents, so the pooled employment estimate averages associations of different sign. These gradients are consistent with substitution where the older generation has time to give, which a fixed household trait would not obviously predict, but they do not rule out selection.

### 4.5 Differences by household arrangement

Table 6 reports the estimates by subsample. For head-level mothers, whose parents or in-laws live in their household, the childcare association is -13.2 minutes (CI -17.3 to -9.2) and the employment association 12.6 minutes (CI 5.6 to 19.5). For junior-level mothers, who live in their in-laws' or parents' household, the childcare association is 3.4 minutes (CI -1.8 to 8.5) and the employment association 7.6 minutes (CI 1.2 to 13.9), estimated from only 2,054 junior-level mothers without a grandmother. The two arrangements are selected for different reasons, the first possibly more often by the grandparents' need for support and the second by the patrilocal norm, and the estimates differ. Because most junior-level mothers live with a grandmother, the pooled grandmother indicator partly captures the difference between the two arrangements. With a control for the mother's generation the estimate is -6.8 minutes (CI -9.9 to -3.6) (see the robustness checks below), compared with -2.1 minutes (CI -4.6 to 0.4) in the pooled specification.

Table 6: Heterogeneity: grandmother coresidence and mothers' childcare and employment minutes

| Subsample | Childcare | Employment | N |
|---|---|---|---|
| Head-level mother (grandparents live in her household) | -13.21 [-17.27, -9.15] | 12.55 [5.59, 19.51] | 32,742 |
| Junior-level mother (lives in the older generation's household) | 3.36 [-1.77, 8.49] | 7.57 [1.21, 13.93] | 17,250 |
| Urban | -6.78 [-11.65, -1.92] | 12.83 [4.98, 20.68] | 16,812 |
| Rural | -0.47 [-3.36, 2.43] | -5.88 [-9.49, -2.27] | 33,180 |
| Mother below secondary education | -1.04 [-4.42, 2.35] | -6.91 [-12.12, -1.70] | 24,804 |
| Mother secondary education or above | -3.52 [-7.04, 0.00] | 5.05 [0.50, 9.60] | 25,188 |
| One child under six | -0.11 [-3.20, 2.99] | 1.30 [-3.88, 6.48] | 26,024 |
| Two or more children under six | -4.35 [-7.99, -0.70] | -2.67 [-6.90, 1.56] | 23,968 |

*Notes:* Main specification, estimated separately by subsample. Estimate [95 percent confidence interval] for the grandmother indicator. Not estimable: fewer than 30 mothers with or without a coresident grandmother in the subsample. No subsample falls below it.

Across urban and rural households the childcare associations are -6.8 minutes (CI -11.7 to -1.9) and -0.5 minutes (CI -3.4 to 2.4). By the mother's education they are -1.0 minutes (CI -4.4 to 2.4) below secondary and -3.5 minutes (CI -7.0 to 0.0) at secondary or above. More educated mothers

have better labor-market options, so if grandmothers' help reaches paid work anywhere it should be among them; the corresponding employment associations are -6.9 minutes (CI -12.1 to -1.7) and 5.1 minutes (CI 0.5 to 9.6), and by urban and rural residence 12.8 minutes (CI 5.0 to 20.7) and -5.9 minutes (CI -9.5 to -2.3). With one child under six the childcare association is -0.1 minutes (CI -3.2 to 3.0), and with two or more -4.4 minutes (CI -8.0 to -0.7). The employment associations differ in sign between urban and rural mothers, which again shows that the pooled employment estimate should not be read as evidence of no association.

### 4.6 A placebo: mothers of school-age children

If grandmothers substitute for the care of young children, the association with mothers' childcare should be weaker where the youngest child is of school age and needs less supervision. If instead the main estimate reflects fixed differences between joint and nuclear households, it should be similar for both groups of mothers. Table 7 reports the comparison for 19,829 mothers whose youngest identified own child is aged six to 14 and who have no child under six in the household. The childcare association is -0.4 minutes (CI -2.3 to 1.6) in this group, against -2.1 minutes (CI -4.6 to 0.4) in the analysis sample, and the difference between the two is -1.8 minutes (CI -4.9 to 1.4). Because the main childcare estimate is itself small, the childcare comparison is not informative on its own. The placebo is not fully clean. The association with domestic work, which the constraint channel predicts should not depend on the child's age, is 8.8 minutes (CI 3.0 to 14.6) in the placebo group and 1.1 minutes (CI -2.2 to 4.5) in the analysis sample, so a grandmother's presence is associated with more domestic work among mothers of school-age children, for whom the substitution channel should be weak. This pattern is consistent with a workload effect of larger households and with selection, and it means the placebo cannot rule out that part of the main estimates reflects household traits.

Table 7: Placebo: grandmother coresidence for mothers whose youngest child is aged 6 to 14

| Outcome | Child under six | Youngest child 6–14 | Difference |
|---|---|---|---|
| Childcare (primary) | -2.09 [-4.58, 0.40] | -0.35 [-2.26, 1.57] | -1.75 [-4.89, 1.39] |
| Active childcare | -2.17 [-4.60, 0.25] | 0.44 [-1.40, 2.28] | -2.61 [-5.66, 0.43] |
| Passive childcare (minding) | -0.05 [-0.56, 0.47] | -0.38 [-0.65, -0.10] | 0.33 [-0.26, 0.91] |
| Unpaid domestic work | 1.14 [-2.19, 4.47] | 8.81 [3.02, 14.61] | -7.68 [-14.36, -0.99] |
| Employment (incl. travel) | -0.41 [-3.79, 2.97] | -0.06 [-7.47, 7.36] | -0.35 [-8.50, 7.80] |

*Notes:* Estimate [95 percent confidence interval] for the grandmother indicator. Column 1: analysis sample (49,992 mothers). Column 2: mothers defined in the same way with no child under six in the household and an identified youngest own child aged 6 to 14 (19,829 mothers). Difference: column 1 minus column 2, with a standard error that ignores the covariance of the two estimates. Controls as in the main specification.

### 4.7 Robustness

Table 8 collects robustness checks for the childcare estimate. Restricting to households with a single married woman aged 18 to 49, where the child under six is most likely hers, gives -3.2 minutes (CI

-5.8 to -0.5). Restricting to head-level mothers gives -13.2 minutes (CI -17.3 to -9.2). Requiring a linked husband gives -1.8 minutes (CI -4.5 to 0.9), and dropping proxy-reported diaries gives -3.4 minutes (CI -6.2 to -0.5). Adding the number of other adult women gives -0.9 minutes (CI -3.4 to 1.6); adding household size gives 8.3 minutes (CI 5.1 to 11.4); and adding an indicator for the mother's generation gives -6.8 minutes (CI -9.9 to -3.6). The unweighted estimate is -1.8 minutes (CI -4.0 to 0.4), and broadening the treatment to any grandparent gives -3.3 minutes (CI -5.7 to -0.8). On the extensive margin, a grandmother is associated with a 1.8 percentage points lower (CI -2.5 to -1.0) probability that the mother reports any primary childcare. The estimates range from -13.2 to 8.3 minutes and do not all share a sign, so the main childcare estimate should be read as fragile: its sign and size depend on whether household size, the mother's generation or the household arrangement is held constant. The bias-adjusted coefficient of Oster (2019), which assumes that unobservables are as important as observables and that the maximum attainable $R^2$ is 1.3 times that of the full model, is -2.33 for childcare and -1.00 for employment minutes. The degree of selection on unobservables, relative to observables, that would reduce the coefficient to zero is -8.66 for childcare and -0.69 for employment; both values are negative, so only selection on unobservables opposite in direction to selection on observables could reduce the coefficients to zero, and for employment the required degree is below one, so the employment estimate is not robust in this sense. Defining the grandmother as a woman aged 45 or above, the lower limit of the older-generation sample, gives -3.1 minutes (CI -5.5 to -0.6) for childcare and 0.6 minutes (CI -2.8 to 3.9) for employment.

Table 8: Robustness: mothers' primary childcare minutes

| Check | Estimate | 95 percent CI |
|---|---|---|
| Baseline | -2.09 | [-4.58, 0.40] |
| Only married woman aged 18–49 in the household | -3.18 | [-5.84, -0.52] |
| Head-level mothers only (head or spouse of head) | -13.21 | [-17.27, -9.15] |
| Mothers with a linked husband only | -1.79 | [-4.46, 0.88] |
| Self-reported diaries only | -3.35 | [-6.22, -0.47] |
| Controlling for other women aged 15+ | -0.94 | [-3.43, 1.56] |
| Controlling for household size | 8.25 | [5.10, 11.39] |
| Controlling for mother's generation (junior) | -6.75 | [-9.94, -3.57] |
| Unweighted | -1.81 | [-4.01, 0.39] |
| Any grandparent instead of grandmother | -3.26 | [-5.73, -0.79] |
| Grandmother defined as a woman aged 45 or above | -3.08 | [-5.54, -0.61] |
| Extensive margin: any primary childcare (share) | -0.018 | [-0.025, -0.010] |
| Oster bias-adjusted, childcare (delta = 1, Rmax = 1.3 R) | -2.33 | |
| Oster bias-adjusted, employment (delta = 1, Rmax = 1.3 R) | -1.00 | |
| Oster delta for a zero coefficient, childcare (Rmax = 1.3 R) | -8.66 | |
| Oster delta for a zero coefficient, employment (Rmax = 1.3 R) | -0.69 | |

*Notes:* Coefficient on the grandmother indicator (or any grandparent) in the main specification, unless stated. The extensive-margin row is a linear probability model. The Oster rows use the short regression without controls or fixed effects on the same sample; the bias-adjusted estimate sets delta = 1, and the delta rows give the relative degree of selection on unobservables that would make the coefficient zero. The age-45 row requires the grandmother of a junior-level mother to be aged 45 or above instead of 40.

# 5 Discussion and conclusion

## 5.1 Interpretation

The pooled association between a coresident grandmother and a mother's primary childcare is small, with a 95% CI of -4.6 to 0.4 minutes a day, and the interval for paid work, -3.8 to 3.0 minutes, also includes zero. These intervals describe one specification. The childcare estimate changes size and sign when the comparison changes (Table 8), and it differs by grandmother and by household arrangement (Tables 5 and 6). The pooled estimate is therefore best read as an average over different associations, not as evidence that grandmothers' care has no effect on mothers' time.

The clearest evidence concerns grandmothers. Where a child under six is present, grandmothers report much more childcare than other older women, and less domestic work, free time and sleep, while their paid work does not fall. Grandmothering in this setting is associated with a substantial transfer of time from older women, whose cost falls on their free time and sleep, while their lower domestic work may reflect tasks shifted to mothers. Children in three-generation households also receive more adult care in total, so the grandmother's care adds to what children receive.

For mothers, three readings are consistent with the pattern, and the evidence allows only a partial ranking. The first is substitution where the grandmother has time. Mothers' childcare is lower where the grandmother is 60 or older, and in a separate split where she is not in paid work, and it is higher where she is under 60 (3.1 minutes (CI 0.1 to 6.1)) or employed (5.8 minutes (CI 2.0 to 9.6)); among head-level mothers the reduction is larger. The age and employment splits are separate regressions, not a joint classification, and with both grandparents present mothers' childcare is not lower. The second is the constraint channel: a larger household and a senior woman's authority raise the mother's domestic load and the wedge on work outside the home, so any childcare time released is absorbed, which fits the higher time on food and meals. The third is selection: mothers who choose, or are placed in, three-generation households differ in ways that also shape their time. Gradients by the grandmother's characteristics are harder to attribute to fixed household traits, but selection remains open: the pooled estimate is sensitive to the specification, and the placebo for domestic work is not clean. Fathers' childcare is also lower with a grandmother present, so part of grandmothers' care appears to replace fathers' contribution rather than mothers'.

Studies in the United States, Europe, Japan and China find that grandparents raise maternal employment (Sasaki 2002; Dimova and Wolff 2011; Maurer-Fazio et al. 2011; Compton and Pollak 2014; Bratti et al. 2018). The pooled estimates here show no such association in India (mothers' paid-work minutes differ little, and their employment rate is 1.4 percentage points lower), although grandmothers provide 37% of the combined childcare of the two women, and the subgroup estimates for paid work differ in sign. The contrast is consistent with the view that the binding constraint on young Indian mothers' employment is not, or not only, the availability of childcare, and with the evidence that household structure and norms restrict married women's work in India (Dhanaraj and Mahambare 2019; Anukriti et al. 2020; Jayachandran 2021), although these data cannot establish

it. The Norwegian finding that formal childcare mainly crowded out informal care (Havnes and Mogstad 2011) is a useful parallel: when one form of care replaces another, the effect on mothers' work depends on what else constrains it.

### 5.2 Implications for research and policy

Policy on childcare and policy on older people's wellbeing are usually considered separately. These results suggest that they interact. Family care is not free: it is provided by older women who report less free time and sleep when a young grandchild is present. As India's population ages (International Institute for Population Sciences et al. 2020; United Nations Population Fund and International Institute for Population Sciences 2023), the supply of grandmother care will depend on older women's health, employment and household position, and the implicit cost to them belongs in any assessment of how families, and public provision, share the care of young children. Expanding formal childcare may matter for mothers in extended households as well as in nuclear ones, although its effect will depend on whether the constraints of the extended household also bind on the use of formal care. All of these statements rest on conditional associations from one cross-section and should be treated with caution.

### 5.3 Limitations

The limitations are those of a single cross-section of diaries. Living arrangements are chosen, and the gradients, the placebo, the comparison of arrangements and the coefficient-stability bounds narrow but do not eliminate the scope for selection. Most junior-level mothers live with a grandmother, so the grandmother indicator is entangled with the household arrangement, and a design that compares mothers within the same arrangement would be preferable. The survey offers no credible instrument for coresidence. One diary day per person measures a day, not a routine (Frazis and Stewart 2012), which adds noise to individual outcomes but does not bias population means. Children under six are counted from the roster and assigned to a mother by a generational rule; the estimate that restricts to households with a single married woman of childbearing age is -3.2 minutes (CI -5.8 to -0.5), but misassignment in households with several junior couples cannot be ruled out. The roster code for parents of the head does not distinguish the head's own parents from the spouse's parents, so for head-level mothers the maternal and paternal grandmother cannot be separated. Grandmother age proxies for health only imperfectly. Finally, the survey does not record which child a care episode is for, so in households with several young children the grandmother's recorded care may serve children other than the mother's. Future rounds that record the recipient of care, and surveys that follow the same families over the birth of a grandchild, would allow sharper tests.

### 5.4 Conclusion

Diaries for every member of the household make it possible to measure directly what grandmothers and mothers do with the day in three-generation households. Identifying the generation above each

mother correctly matters, because in patrilocal households the young mother lives in her parents-in-law's household and, unless the roster is read carefully, the head's wife would be counted as a mother rather than as the grandmother she is. With that identification, the pooled association between a coresident grandmother and mothers' childcare or paid-work minutes is small and not robust, while grandmothers' own childcare is substantial and is associated with less free time and sleep for them, and with less domestic work that mothers may partly take over. Family care in India is best understood as a transfer of time between women of two generations whose terms depend on who the grandmother is and on how the household is arranged, and whose cost falls on older women's rest and leisure.

## Statements and Declarations

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**Competing interests.** The author has no relevant financial or non-financial interests to disclose.

**Ethics approval and consent to participate.** The study is a secondary analysis of anonymized, publicly released unit-level microdata from the Time Use Survey 2024 of the Government of India's National Statistics Office. It involved no contact with respondents, so ethics approval and consent to participate were not required. The survey agency obtained informed consent from respondents when it collected the data.

**Consent for publication.** Not applicable.

**Data availability.** The Time Use Survey 2024 unit-level data are available from the microdata portal of the Ministry of Statistics and Programme Implementation under its terms of use. Analysis code will be made available on acceptance.

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